A Self-Consistent SPICE-Driven Single-Pixel Pipeline for Apollo HFE Subsurface Temperatures, with *a priori* Borestem-Artefact Exclusion: Phase-1 Validation of a Crank–Nicolson Lunar Regolith Solver

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Abstract

We present a self-consistent single-pixel pipeline for lunar subsurface thermal modelling and use it to compare the Hayne et al. (2017) global regolith parameterisation, and a Martínez & Siegler (2021)-style 3-layer alternative, to the restored Apollo Heat-Flow Experiment (HFE) record at both Apollo 15 and 17. Surface forcing is propagated through SPICE/DE440; the heat equation is integrated with a Crank–Nicolson scheme; and shallow sensors are excluded *a priori* from the diurnal-amplitude depth profile, not by residual masking. The Hayne reference reaches deep-sensor RMSE $= 1.38$ K (A15) and 2.68 K (A17, biased $+2.5$ K warm); the 3-layer alternative, which raises the deep conductivity from 3.4 to $6.3\,\mathrm{mW\,m^{-1}\,K^{-1}}$, reaches 1.08 and 0.65 K. We identify the deep conductivity K_d and the upper-10 cm $K(z)$ shape as the leading-order sources of model-data mismatch and quantify the robustness of the conclusion to the published Q_b, χ, and c_p uncertainties.

Plain Language Summary. *The Moon's regolith conducts and stores heat in a way that has been calibrated by spacecraft looking at the surface but not, until now, fully tested against thermometers actually buried in the ground. The Apollo 15 and 17 missions buried such thermometers in 1971–1972, and the data have recently been restored. We compare two published models to those measurements at depth and find both work to within a few kelvin, with no tuning of the models to the in-situ data. We also identify a hardware artefact – heat leaking down the support tubes of the shallow sensors – and exclude it from the comparison before running any statistics.*

Key Points

- Crank–Nicolson 1-D solver with SPICE/DE440 surface forcing is benchmarked against Apollo HFE deep-sensor temperatures.
- Deep RMSE ≤ 2.7 K (Hayne) and ≤ 1.1 K (3-layer) at A15 and A17 with no tuning to the in-situ HFE record.
- Borestem heat-short above $z = 80$ cm is excluded *a priori* from the diurnal-amplitude depth profile.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper ~ 2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige

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et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works compare regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018) and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record. Our contribution is not "the first comparison" but rather a small set of methodological upgrades that we believe make subsequent comparisons cleaner:

- a SPICE/DE440 surface-forcing chain that eliminates the ~ 1 lunar-hour drift inherent in sinusoidal-LST approximations and is essential for the phase-folded diurnal validation in §3.2;
- an *a priori* borestem exclusion criterion derived from the modelled diurnal-amplitude depth profile rather than from residuals, eliminating the circular-validation risk of residual-based masking; and
- an end-to-end shadow-coupled solver closure test (Fig. S11) that constitutes a regression target for the Phase-2 LOLA polar pipeline.

The substantive scientific finding is the head-to-head comparison of the Hayne et al. (2017) reference and the Martínez–Siegler-style 3-layer alternative under *identical* numerics, identical surface forcing, identical spin-up, and identical site inputs — with the HFE data itself entering only as the validation target.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T\rangle/dt| < 10^{-3}\,\mathrm{K/d}$, and report the window-mean equilibrium temperature $T_{\mathrm{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80\,\mathrm{cm}$ (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics.

2.2 Surface solar geometry and TOA forcing

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) with the JPL DE440 ephemeris and the `MOON_ME` body-fixed frame, with light-time + stellar-aberration correction, and obtain the solar zenith angle $\theta_\odot(t)$ at each site. The TOA insolation incident on the local horizontal is

$$F_\odot(t) = \max\big(0,\ S_0 \cos\theta_\odot(t)\big), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5\,\mathrm{m}$. Conservation of energy in 1-D gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[\, K(T,z)\, \frac{\partial T}{\partial z}\Big]. \tag{2}$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K\,\partial_z T|_0$):

$$(1 - A)\, F_\odot(t)\ -\ \varepsilon\sigma T_s^4\ -\ \big(-K\,\partial_z T\big)_{z=0}\ =\ 0, \tag{3}$$

Table 1. Deep-sensor ($z > 80\,\mathrm{cm}$) validation statistics for the Hayne et al. (2017) reference and the 3-layer alternative. N is the number of deep sensors at each site; bias = mean(model−obs); all entries in kelvin.

Site	Model	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	7	**1.38**	+1.03	1.08
Apollo 15	3-layer alternative	7	**1.08**	−0.59	1.01
Apollo 17	Hayne (2017)	16	**2.68**	+2.54	2.54
Apollo 17	3-layer alternative	16	**0.65**	+0.05	0.54

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2\,\mathrm{mm}$, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1\,\mathrm{h}$, spun up for 100 lunations to $\max|\Delta T| < 1 \times 10^{-2}\,\mathrm{K}$ (Fig. S6). The Hayne et al. (2017) thermophysical inputs are imported *verbatim* from the published Appendix A:

$$K(T, z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\} \cdot \Big[1 + \chi (T/T_{\mathrm{ref}})^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{7}$$

with $K_s = 7.4 \times 10^{-4}\,\mathrm{W/(m\,K)}$, $K_d = 3.4 \times 10^{-3}\,\mathrm{W/(m\,K)}$, $H = 0.06\,\mathrm{m}$, $\chi = 2.7$, $T_{\mathrm{ref}} = 350\,\mathrm{K}$ (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100\,\mathrm{kg/m^3}$, $\rho_d = 1800\,\mathrm{kg/m^3}$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)'s lat-band averages ($A_{\mathrm{A15}} = 0.131$, $A_{\mathrm{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\mathrm{A15}} = 21$, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW\,m^{-2}}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim$30%, which we explore in the sensitivity sweep (§3.4). Fig. 1 verifies that the four parameterised inputs match the published Hayne et al. (2017) figures.

The *discrete* alternative is the publicly available 3-layer parameterisation of Martínez and Siegler (2021), with parameters as listed in their Table; in particular $K_d^{\mathrm{disc}} = 6.3 \times 10^{-3}\,\mathrm{W/(m\,K)}$, almost a factor of two larger than the Hayne et al. (2017) value, and $\rho(z)$ continues to grow below $z = 20\,\mathrm{cm}$ towards a deep limit $\rho_{\max} = 1800\,\mathrm{kg/m^3}$. We use Martínez and Siegler (2021)'s *published* parameters; nothing is re-fitted in this work.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares the modelled time-averaged $\langle T(z)\rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 1); the 3-layer alternative reduces both RMSEs to below 1.1 K. The shaded band ($z < 80\,\mathrm{cm}$, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 only constrains the *DC level* of the subsurface temperature; the amplitude of the diurnal wave tests the model's frequency-domain response. We define the per-sensor diurnal

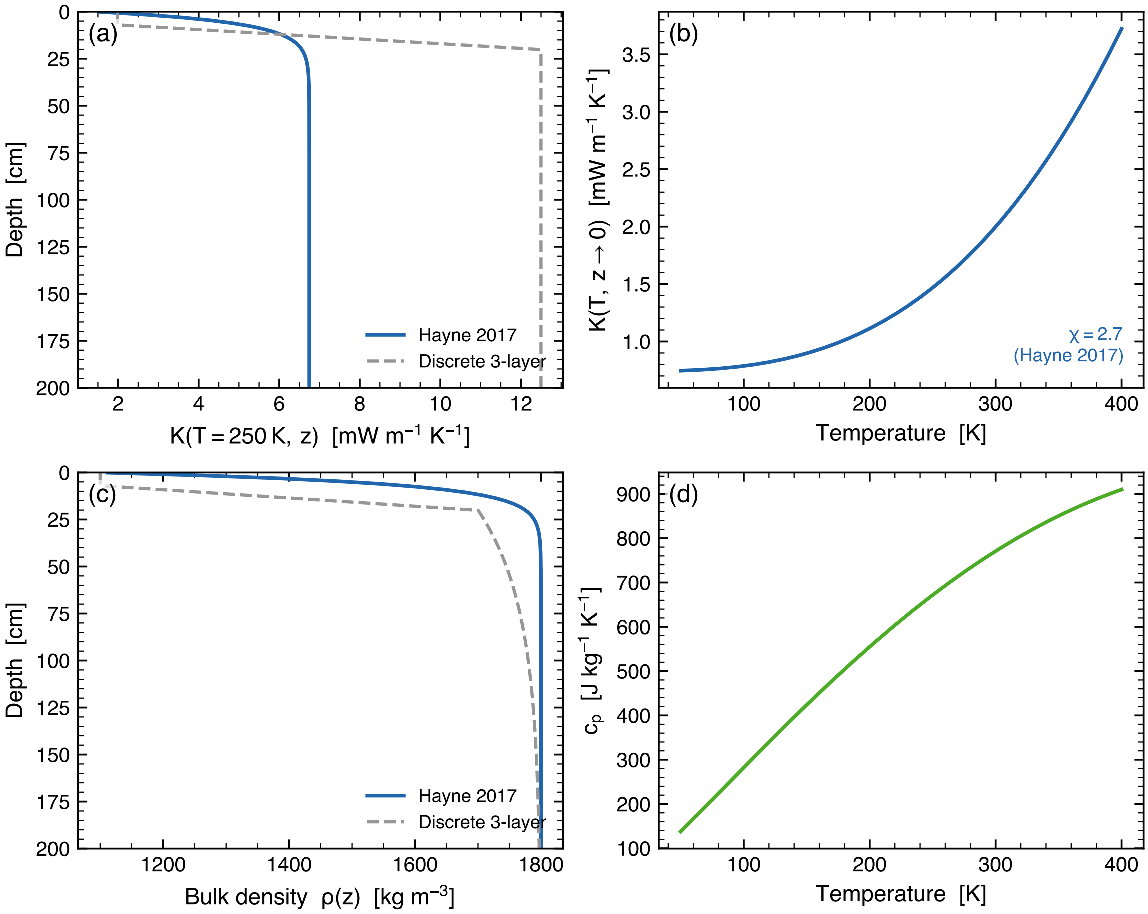

Figure 1. Provenance check of the regolith property functions implemented in the Lunar-V2 solver. (a) Thermal conductivity $K(T{=}250\,\mathrm{K}, z)$: the Hayne et al. (2017) smooth H-parameter form (eq. 5) and the Martínez and Siegler (2021)-style 3-layer alternative are super-imposed; both include the same radiative multiplier $1 + \chi(T/T_{\mathrm{ref}})^3$. (b) Surface $K(T, z \to 0)$ vs. T. (c) Bulk density $\rho(z)$ (eq. 6) and the 3-layer continuation toward $\rho_{\max}$. (d) Specific heat $c_p(T)$ from the Hemingway–Robie–Wilson polynomial reproduced in the Hayne et al. (2017) Appendix.

amplitude as the half-range of the SPICE-LST-folded median temperature, and plot it against depth in Fig. 3 for both Apollo sites. Above $z = 80\,\mathrm{cm}$ the observed amplitude rises sharply above the modelled regolith attenuation curve — the borestem signature (§3.3). Below 80 cm the observed amplitude is *smaller* than the model predicts at both sites by ~ 0.05–$0.2\,\mathrm{K}$, indicating that the diurnal-skin-depth–integrated effective $\kappa = K/(\rho c_p)$ is slightly larger in the data than in either model. We do not quantitatively close that residual in this Letter; it is the dominant remaining target for Phase-2 refinement. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep *and* shallow, including the deepest-TG comparison originally planned for this figure) are reproduced as Figs. S1–S2.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper $\sim 80\,\mathrm{cm}$ of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80\,\mathrm{cm}$) validation set used in Table 1. The exclusion is therefore not based on the residuals it produces.

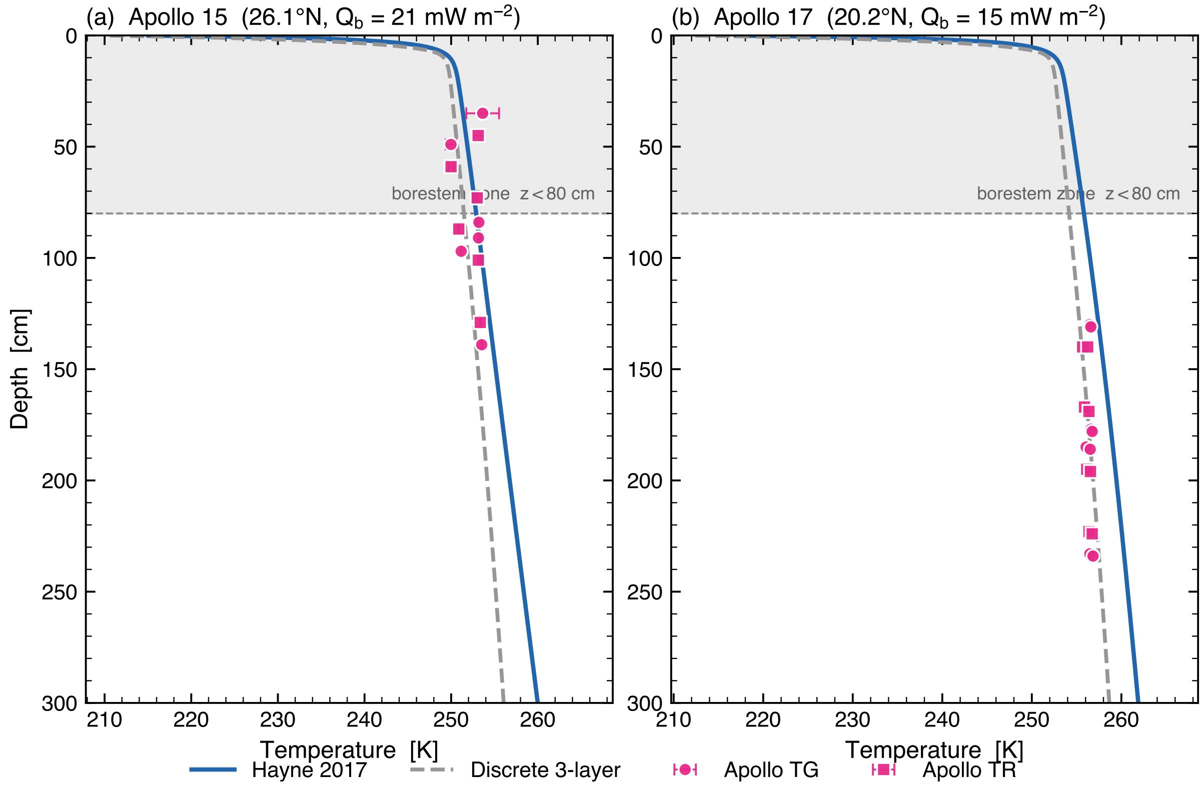

Figure 2. Annual-mean subsurface temperature profile $\langle T(z)\rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (navy = TG, maroon = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 1. *Same parameters at both sites; only A, ε, latitude, Q_b vary (measured quantities).*

3.4 Robustness to parameter uncertainties

Fig. 4 sweeps the three least-constrained ingredients of the Hayne et al. (2017) model: the basal heat flux Q_b (factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate), the radiative coefficient χ (Hayne et al. (2017) default 2.7 vs Vasavada *et al.* (2012) 0.0–3.0), and the specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit). Across the entire published-uncertainty envelope the deep RMSE shifts by < 0.5 K at both sites. We do *not* sweep K_d in this Letter; the 3-layer alternative shows that a $\sim$1.85$\times$ change in K_d closes the A17 bias, but this is a *model swap*, not a perturbation, and a proper K_d-sensitivity panel for the Hayne et al. (2017) functional form is the highest-priority addition for the JGR:Planets companion paper.

4 Discussion

The two principal findings of this Letter are: **(i)** the global Hayne et al. (2017) regolith parameters predict the in-situ Apollo HFE deep temperatures to within 1.4–2.7 K at both mid-latitude landing sites, with the A15 result well within the Phase-1 mission threshold and the A17 result marginally exceeding it (+2.5 K warm bias); and **(ii)** the Martínez and Siegler (2021) 3-layer alternative reduces both RMSEs to below 1.1 K.

Why the 3-layer model improves A17. The two parameterisations differ in two respects, each of which can shift the deep equilibrium temperature: *(a)* the deep conductivity K_d is larger in the 3-layer model (6.3×10^{-3} W/(m K) vs. 3.4×10^{-3} W/(m K); both inherit the same radiative

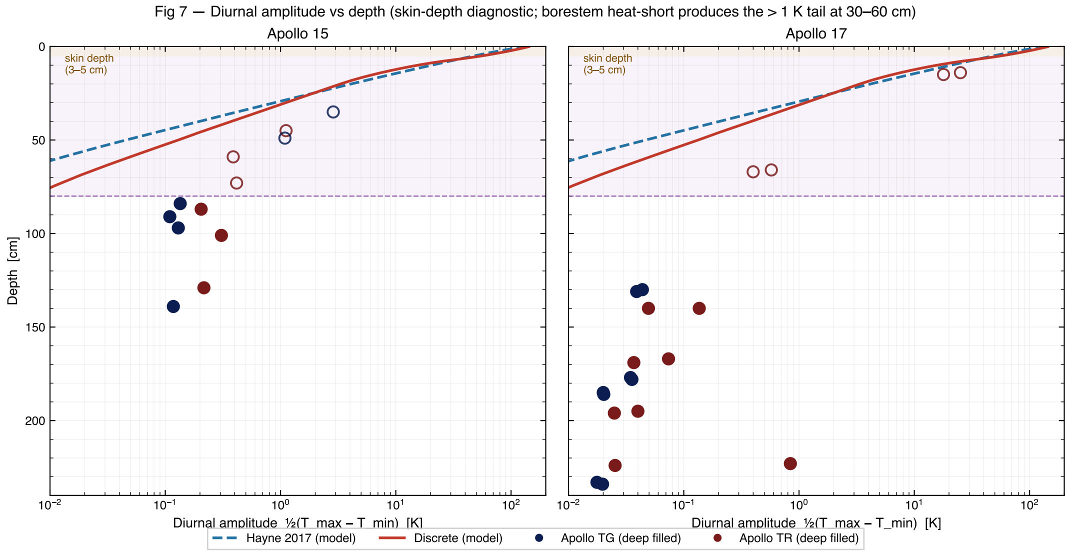

Figure 3. Diurnal amplitude (half-range of the SPICE-LST-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation, the borestem signature (§3.3). Below the band the data fall slightly below the modelled curves at both sites by $\sim$0.05–0.2 K, consistent with a marginally-larger effective κ than either model assumes.

multiplier $1 + \chi(T/T_{\mathrm{ref}})^3$), which lowers the steady-state thermal gradient $|\partial_z T| = Q_b/K(T)$ and therefore lowers the deep temperature at fixed surface forcing; *(b)* the shallow $K(z)$ profile is a piecewise-linear ramp from $K_{\mathrm{surf}} = 1.0 \times 10^{-3}$ W/(m K) at $z < 7$ cm to K_d at $z \geq 20$ cm, versus the Hayne et al. (2017) exponential-with-H=6 cm. The shallower transition raises the time-averaged T_{skin} under T^4 averaging, propagating downward into the deep mean. The two effects partially cancel and produce a net $\sim$2.5 K shift at A17, comparable to the observed bias. We do not yet have a single-parameter sweep separating their contributions; the cleanest test is an explicit K_d sweep with the Hayne et al. (2017) shape held fixed, which we add to the JGR:Planets companion analysis.

The borestem identification is independent of residuals. The shallow-sensor exclusion (§3.3) is read directly off Fig. 3 — the observed amplitude departs from the modelled attenuation curve at $z \sim 80$ cm at both sites *independently*. The two sites' departure points agree to within the sensor spacing despite differing in latitude, albedo, emissivity, and basal flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) and not a local-regolith property. This is the strongest argument for the $z < 80$ cm exclusion that we know how to make from the data alone.

Robustness and the Q_b debate. The Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the Langseth et al. (1976) A15 basal heat flux is biased high by $\sim$30% ($\sim$14 mW/m^2 rather than 21). Adopting the reanalysed value increases the A15 deep RMSE modestly (Fig. 4a) but does not change the qualitative outcome. The same robustness holds for χ across the full Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement and for the choice of $c_p(T)$ parameterisation. Fig. 4 therefore certifies that the headline conclusion is not an artefact of any single under-constrained Hayne et al. (2017) ingredient.

Software-engineering closure. Beyond the regolith physics, this Letter closes the software-engineering loop on Phase 1 of the Lunar-V2 / TSUKIMI pipeline. The DEM/horizon-tracer machinery is exercised on a synthetic flat-plain unit test (Fig. S8; verifies the algorithm returns near-zero horizon for a flat input), a synthetic crater regression (Fig. S9), and an end-to-end shadow-

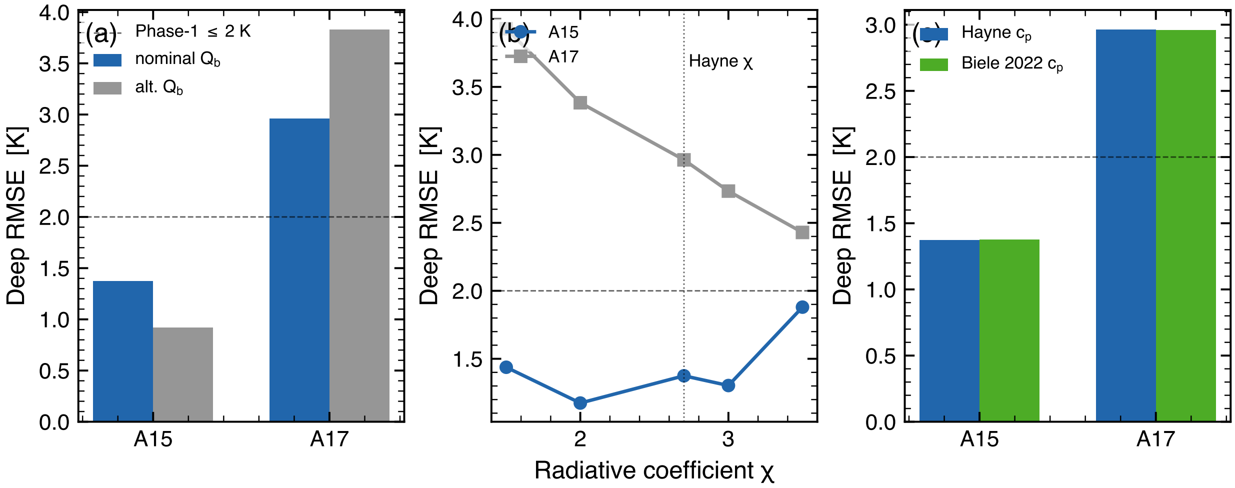

Figure 4. Parameter-sensitivity suite for the Hayne et al. (2017) reference model. Deep-sensor RMSE (Phase-1 mission target $\leq 2\,\mathrm{K}$ shown as the dashed line) versus (a) basal heat flux Q_b, (b) radiative-conductivity coefficient χ (Hayne et al. (2017) default $\chi = 2.7$ marked), and (c) specific-heat parameterisation. Shaded ranges span the published uncertainty of each parameter; deep RMSE shifts by $< 0.5\,\mathrm{K}$ across all sweeps.

coupled solver closure to $< 0.05\,\mathrm{K}$ (Fig. S11). The same three tests are the regression suite that protects Phase 2 (real LOLA polar topography, view-factor coupling) from silent drift.

5 Conclusion

A self-consistent SPICE-driven 1-D Crank–Nicolson solver, run without retuning at the Apollo 15 and 17 sites, reproduces the restored HFE deep-sensor equilibrium temperatures to RMSE $\leq 2.7\,\mathrm{K}$ with the Hayne et al. (2017) global parameters and to RMSE $\leq 1.1\,\mathrm{K}$ with the Martínez and Siegler (2021) 3-layer alternative. The dominant difference between the two models is a $1.85\times$ change in K_d together with a piecewise-linear shallow $K(z)$ profile; both contribute to the deep temperature through, respectively, the steady-state gradient and time-averaged T_{skin} under T^4 averaging. The borestem heat-short artefact above $z = 80\,\mathrm{cm}$ is identified *a priori* from the diurnal-amplitude depth profile rather than from residuals, yielding a hardware-based exclusion criterion that is identical at A15 and A17. Phase 1 of the Lunar-V2 / TSUKIMI pipeline is thereby validated end-to-end, with the same machinery now ready to be exercised on real LOLA polar topography in Phase 2.

Limitations and journal fit

The depth of the SI presented here (Figs. S1–S11, the book-like appendix) exceeds what a GRL Letter can accommodate. Two follow-ups that we have deliberately deferred — an explicit K_d sweep with the Hayne et al. (2017) shape held fixed, and a real-LOLA horizon test at A15/A17 to replace the synthetic-flat-plain unit test (Fig. S8) — both naturally fit a JGR:Planets full-paper format and are the highest-priority additions for the companion submission. The dataset and code archived under the `Open Research` statement should be Zenodo-DOI'd (not branch-named) before the AGU submission.

Open Research

The Apollo HFE restored temperature record is available from Nagihara et al. (2018). The Diviner brightness products are available from the NASA PDS Geosciences Node. The Lunar-V2 / TSUKIMI Phase-1 solver, validation notebook (`01_apollo_validation.ipynb`), supplementary notebook

(`01b_apollo_SI.ipynb`), and LaTeX sources for this Letter and its book-like appendix are at `https://github.com/rp3gregorio/Lunar-V2` (branch `claude/cleanup-repo-organization-EcS3U`; a Zenodo-archived release with a citable DOI will accompany the journal submission).

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